Gradient descent is a fundamental iterative optimization algorithm used in machine learning to minimize a model's error (cost function) by adjusting its parameters (e.g., weights and biases). It works by calculating the steepest downhill direction (the gradient) and taking small steps in the opposite direction, converging towards the minimum value.

Key Concepts of Gradient Descent:

- **Purpose:** To find the optimal parameters that minimize the loss/cost function, reducing prediction errors
- **The Gradient:** A vector of partial derivatives that indicates the direction of steepest ascent; therefore, moving in the *negative* gradient direction leads to the minimum.
- **Learning Rate:** A hyperparameter that determines the size of the steps taken towards the minimum. A rate too small makes training slow, while too large may cause it to overshoot the minimum.
- **Process:** Starts with random parameter values and iteratively updates them:

Main Types of Gradient Descent:

- **Batch Gradient Descent:** Computes the gradient using the entire dataset, which is stable but slow on large data.
- **Stochastic Gradient Descent (SGD):** Updates parameters using only one training sample at a time, making it faster but more volatile.
- **Mini-Batch Gradient Descent:** A hybrid approach that updates using small subsets of the data, balancing speed and stability.